Signals and Systems

Grade Test: Quiz 3

Test Information

QUESTION 1: CALCULATED FORMULA

5 **out of 5 points**

If X3=6+8j, what is the value of X_C3?

Given Answer: 12

Correct Answer: 12

QUESTION 2: MULTIPLE CHOICE | 15 | out of 15 points

The signal x(t)=rect(20t)*[10comb(10t)] is applied to the input of a filter whose transfer function is given by tri(f/30). Which one of the below signals corresponds to the output signal y(t)?

Given Answer: $\frac{1}{2}+\frac{4}{3\pi}\cos(20\pi t)$

Correct Answer: $\frac{1}{2}+\frac{4}{3\pi}\cos(20\pi t)$

QUESTION 3: MULTIPLE CHOICE | 10 | out of 10 points

Consider a real signal of period 0.01 sec whose FS coefficients are given by X[0]=2 and X[k]=6/|k| when k is different from zero. This signal is applied to the input of an ideal bandpass filter and the output signal is given by $6\cos(400\pi t)+4\cos(600\pi t)+3\cos(800\pi t)$. Which one of the below frequencies corresponds to a valid option of the lower cutoff frequency (in Hz)?

Given Answer: 150

Correct Answer: 150

QUESTION 4: CALCULATED FORMULA | 5 | out of 5 points

Consider a real signal x(t) whose nonzero FS coefficients (for k>=0) are given by:

- X0=5
- X1=1+3j

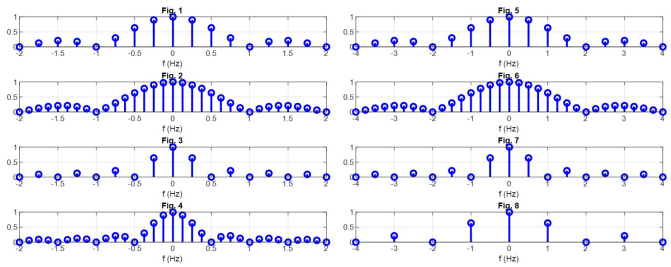
What is the average power of the signal?

Given Answer: 45

Correct Answer: 45

QUESTION 5: MULTIPLE CHOICE | 15 | out of 15 points

Which one of the below figures corresponds to the magnitude spectrum of the signal $x(t) = \text{rect}(t) * \text{comb}\left(\frac{t}{2}\right)$?



Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 6: MULTIPLE CHOICE

10 out of 10 points

Consider the signal given by $x(t) = 2\cos(200\pi t) - 4\sin(600\pi t)$. What are the nonzero FS coefficient X[k] (for k>0)?

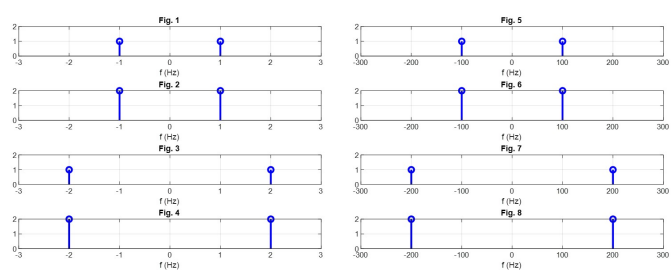
Given Answer: X[1]=1 and X[3]=2j

Correct Answer: X[1]=1 and X[3]=2j

QUESTION 7: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the magnitude spectrum of the signal $2\sin(400\pi t + \pi/6)$?



Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 8: CALCULATED FORMULA 10 **out of 10 points**

What is the average power of the signal 2rect(t/3)*comb(t/8)?

Given Answer: 96

Correct Answer: 96

QUESTION 9: MULTIPLE CHOICE 5 **out of 5 points**

Consider a signal whose nonzero FS coefficients are given by:

- X[-3]=j
- X[3]=-j
- X[-5]=-4j
- X[5]=4j

Which one of the below statements best describes the signal?

Given Answer: The signal is real and odd.

Correct Answer: The signal is real and odd.

QUESTION 10: MULTIPLE CHOICE 10 **out of 10 points**

Consider a real signal of period 0.2 sec whose FS coefficients are given by X[0]=2, $X[k] = \frac{1}{k^2}$ when k is odd and X[k]=0 otherwise.

This signal is applied to the input of an ideal lowpass filter of gain G. The output signal is given by $4 + a\cos(10\pi t)$. What are the values of G and a?

Given Answer: G=2 and a=4

Correct Answer: G=2 and a=4

QUESTION 11: MULTIPLE CHOICE 5 **out of 5 points**

Consider the signal x(t) given by:

$x(t) = -5 + 2\cos(50\pi t) + 3\sin(100\pi t) + \cos(150\pi t) + \sin(350\pi t)$

When x(t) is applied to the input of a filter, the output signal is given by:

$$y(t) = -5 + 2\cos(50\pi t) + \cos(150\pi t)$$

What is the type of the filter?

Given Answer: None of the above.

Correct Answer: None of the above.

FEEDBACK AND NOTES FOR ATTEMPT

Feedback to Learner

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0

Grading Notes

Paragraph Arial 3 (12pt)

Mashups

Path: p Words:0